



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

PRE - SCORE

17-11-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : I(A), TTAS & TEAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	B	D	B	D	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	B,C,D	A,B,C	B,C	B	A,B,C	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	6.00	25.82	5.50	1.73	0.33	3.80

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	D	C	A	C	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,D	A,C,D	A,B,D	B,C,D	A,C,D	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	3.00	500.00	4.00	5.00	5.00	2.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	A	A	A	C	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	C,D	C,D	A,D	B,C,D	B,C	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.00	5.00	9.00	7.00	9.00	6.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (A)**

$$F_{\text{net}} = 9\text{N}$$

$$\tau_{\text{act}} = 6 \times 2 - 3 \times 1 = 9 = F_{\text{net}} \cdot x$$

$$\therefore x = 1\text{m}$$

2. **Ans (B)**

$$u = 10 - 2t$$

$$v = 2t$$

$$v = \frac{uF}{u + F} \Rightarrow 2t = \frac{-(10 - 2t) 0.9}{- [10 - 2t] + 0.9}$$

$$\Rightarrow 2t [-9.1 + 2t] = [-10 + 2t] 0.9$$

$$\Rightarrow -18.2t + 4t^2 = -9 + 1.8t$$

$$4t^2 - 20t + 9 = 0$$

$$t = \frac{20 \pm \sqrt{(20)^2 - 4(4) \times 9}}{2 \times 4}$$

$$t = \frac{20 \pm \sqrt{400 - 144}}{8}$$

$$t = \frac{20 \pm 16}{8} \Rightarrow t_1 = \frac{4}{8} = \frac{1}{2} = 0.5 \text{ sec}$$

$$\Rightarrow t_2 = \frac{36}{8} = \frac{9}{2} = 4.5 \text{ sec}$$

3.

Ans (D)

Since collision is elastic $\therefore$ Velocity of

dielectric after collision is $\frac{2v_0}{3}$

Dielectric will move and when it is coming out of capacitor a force is applied on it by the

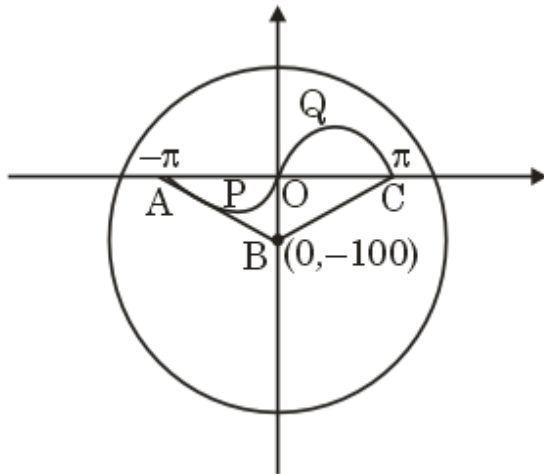
$$\text{capacitor } F = -\frac{dU}{dx} = -\frac{E^2 \epsilon_0 b (K-1)}{2d}$$

Which decreases its speed to zero, till it comes out it travels a distance a

$$\frac{4}{9} \times 2 \frac{1}{2} M v_0^2 = \frac{E^2 \epsilon_0 b (K-1) a}{2d}$$

$$v_0 = E \left[\frac{\epsilon_0 a b (K-1)}{2Md} \right]^{1/2} \times \frac{3}{2}$$

4. Ans (B)



Connect centre B with the two end points A & C of the curves, by conducting rods.

∴ Electric lines of force will be perpendicular to AB & CB.

∴ E.M.F. developed in the loop, BAPOQCB will be = E.M.F. developed in the curve APOQC.

Now Flux in loop ABCQOPA = B

[Area of ΔABC - Area of loop APOA + Area of loop OCOA] = B [Area of ΔABC]

$$= \frac{1}{2} \times (100) \times (2\pi)$$

$$= 100\pi = 314 \text{ m}^2 \therefore \phi = 314 \text{ B}$$

$$\therefore \varepsilon = 314 \frac{dB}{dt} = 314 \text{ V}$$

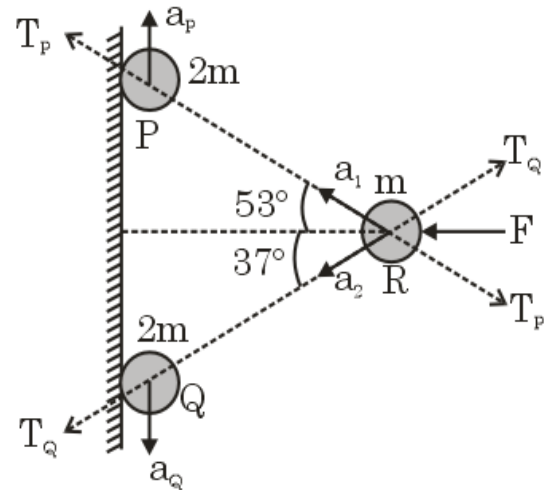
5. Ans (D)

$$I = \left(\frac{58}{1000} \right) (64) = \text{Area}$$

$$\frac{(58)(64)}{1000} = \frac{1}{2} \frac{(8)}{1000} (F_{\max})$$

$$\Rightarrow F_{\max} = 928 \text{ N}$$

6. Ans (C)



Let component of acceleration of disc R are

a_1 and a_2

$$a_1 = a_P \cos 37^\circ = \frac{4a_P}{5} \dots (1)$$

$$a_2 = a_Q \cos 53^\circ = \frac{3a_Q}{5} \dots (2)$$

For P

$$T_P \cos 37^\circ = 2ma_P$$

$$\frac{4T_P}{5} = 2ma_P \dots (3)$$

For Q

$$T_Q \cos 53^\circ = 2ma_Q$$

$$\frac{3T_Q}{5} = 2ma_Q \dots (4)$$

For R

$$F \cos 53^\circ - T_P = ma_1$$

$$\frac{3F}{5} - T_P = ma_1 \dots (5)$$

$$\text{and } F \cos 37^\circ - T_Q = ma_2$$

$$\frac{4F}{5} - T_Q = ma_2$$

from (3) & (5)

$$\frac{3F}{5} - \frac{10}{4} ma_P = ma_1$$

using (1) in above relation we get

$$a_P = \frac{2F}{11m}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (B,C,D)**

$$(A) A \cos \omega t = \left(\frac{1 - \cos 2\omega t}{2} \right)$$

$$= \frac{A}{2} [\cos \omega t - \cos \omega t \cos 2\omega t]$$

$$= \frac{A}{2} \left[\cos \omega t - \frac{1}{2} \cos \omega t + \frac{1}{2} \cos 3\omega t \right]$$

$\therefore$ Not SHM

11. **Ans (A,B,C)**

$$(A) eV = \frac{1}{2}mv_0^2 \text{ \& } \lambda = \frac{h}{mv_0} = \frac{h}{\sqrt{2emV}}$$

$$(B) \beta = \frac{\lambda D}{d} \therefore \beta \propto \lambda \propto \frac{1}{\sqrt{V}}$$

$$(C) N_{\max} = 2 \left[\frac{d}{\lambda} \right] + 1 > 3 \Rightarrow \left[\frac{d}{\lambda} \right] > 1 \text{ or } d > \lambda$$

$$\Rightarrow d^2 > \frac{h^2}{2emV^2} \therefore V^2 > \frac{h^2}{2emd^2}$$

PART-1 : PHYSICS

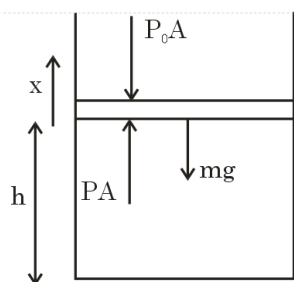
SECTION-II

1. **Ans (6.00)**

$$N = mg \cos \theta$$

$$mg \sin \theta + m\omega^2 r = \mu N$$

2. **Ans (25.82)**



$$PA = P_0A + mg \dots\dots(1)$$

$$P'A - (P_0A + mg) = \frac{md^2x}{dt^2} \dots\dots(2)$$

$$P(Ah)^\gamma = P'A(h+x)^\gamma \dots\dots(3)$$

$$\therefore P' \approx P \left(1 - \frac{\gamma x}{h} \right)$$

$$PA \left(1 - \frac{\gamma x}{h} \right) - P_0A = m \frac{d^2x}{dt^2}$$

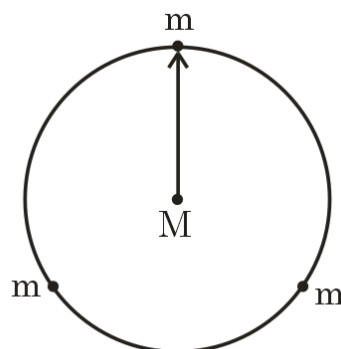
$$\Rightarrow \frac{d^2x}{dt^2} = - \frac{\gamma PA}{mh} x$$

$$\therefore \omega = \sqrt{\frac{\gamma (P_0A + mg)}{mh}}$$

3. **Ans (5.50)**

$$\frac{hc}{\lambda} = \phi + k_{\max} = \phi + CV_0 = \phi + e \frac{kq_{\max}}{R}$$

4. **Ans (1.73)**



$$U = -\frac{GMm}{R} \times 3 - \frac{Gm^2}{\sqrt{3}R} \times 3$$

$$= -2\sqrt{3} \frac{Gm^2}{R} = -\sqrt{3} \times 10^{26}$$

5. **Ans (0.33)**

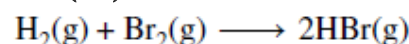
$$d = 0.5 \text{ g t}^2 \Rightarrow \ell \text{nd} = \ell \text{n } 0.5 \text{ g} + 2 \ell \text{nt}$$

$$\frac{d(d)}{d} = 2 \frac{dt}{t} = \frac{2 \times 0.5}{3}$$

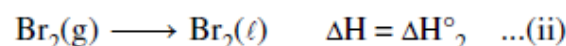
PART-2 : CHEMISTRY

SECTION-I (i)

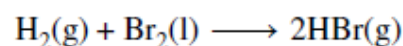
2. **Ans (D)**



$$\Delta H = \Delta H^\circ_1 \dots\dots(i)$$

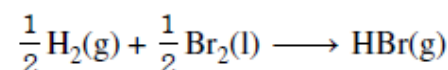


$$[eq_1 - eq_2]$$



$$\Delta H = \Delta H^\circ_1 - \Delta H^\circ_2$$

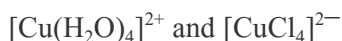
Required equation,



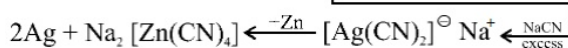
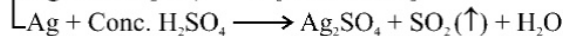
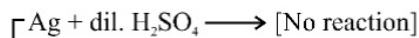
$$\Delta H = \left[\frac{\Delta H^\circ_1 - \Delta H^\circ_2}{2} \right]$$

3. **Ans (C)**

Due to formation of both complex.



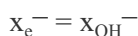
4. **Ans (A)**



PART-2 : CHEMISTRY

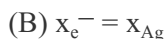
SECTION-I (ii)

8. **Ans (A,C,D)**



$$1 \times \frac{9.65 \times 100}{96500} = 1 \times n_{\text{OH}^-} \Rightarrow [\text{OH}^-] = 10^{-2}$$

$$\Rightarrow \text{pOH} = 2 \Rightarrow \text{pH} = 12$$

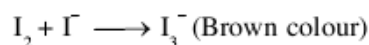
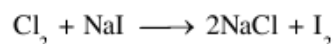
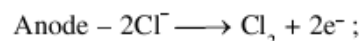
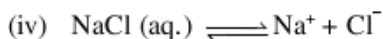
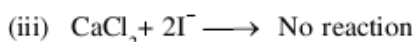
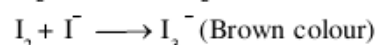
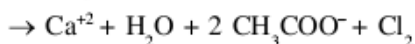
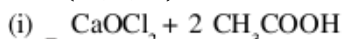


$$1 \times \frac{9.65 \times 100}{96500} = \frac{m_{\text{Ag}}}{108} ; m_{\text{Ag}} = 1.08\text{gm}$$



(D) In electrolytic cell B and electrolytic cell C oxidation of water take place at anode so both have equal pH after electrolysis.

9. **Ans (A,B,D)**

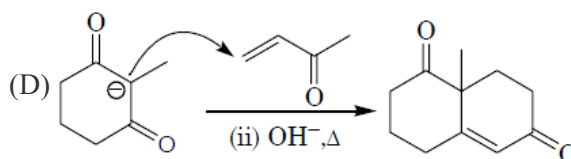


11. **Ans (A,C,D)**

(A) Aromatic chemistry.

(B) $\text{Nu}^\ominus$ addition

(C) Ozonolysis following Aldol condensation.



12. **Ans (B,C,D)**

Theory based.

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (3.00)**

$$\ln \left[\frac{(t^{1/2})_1}{(t^{1/2})_2} \right] = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\ln \left[\frac{3}{(t^{1/2})_2} \right] = \frac{840R}{R} \left(\frac{1}{300} - \frac{1}{400} \right)$$

$$(t^{1/2})_2 = 1.5 \text{ min.}$$

2. **Ans (500.00)**

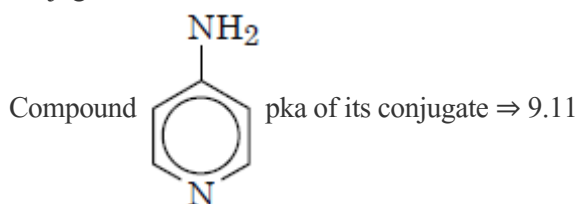
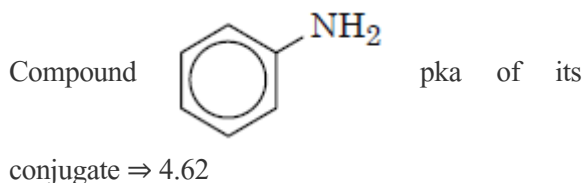
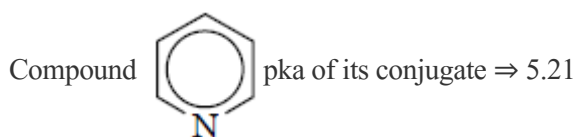
$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$0 = \Delta H^\circ - T\Delta S^\circ$$

$$\Rightarrow T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{25 \times 10^3}{50} = 500\text{K}$$

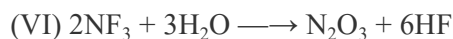
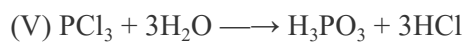
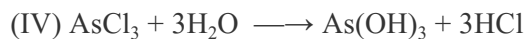
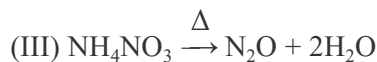
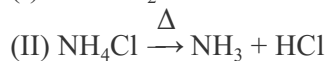
3. **Ans (4.00)**

(A), (C), (E) and (F) are correct.



(G) % of s-character.

6. Ans (2.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (D)

Let $a = \sin^2 x$ and $b = \cos^2 x$. Since

$$\frac{1}{2} \left(\frac{1}{a} + \frac{1}{b} \right) \geq \left(\frac{2}{a+b} \right) \text{ and } a+b=1 \text{ we}$$

$$\text{have } \left(\frac{1}{a} + \frac{1}{b} \right) \geq 4$$

where equality holds only when $a = b = 1/2$.

Therefore

$$\begin{aligned} 12 + \frac{1}{2} \sin y &= \left(a + \frac{1}{a} \right)^2 + \left(b + \frac{1}{b} \right)^2 \\ &\geq \frac{1}{2} \left\{ \left(a + \frac{1}{a} \right) + \left(b + \frac{1}{b} \right) \right\}^2 \\ &= \frac{1}{2} \left(a + b + \frac{1}{a} + \frac{1}{b} \right)^2 \geq \frac{1}{2} (1 + 4)^2 = \frac{25}{2} \end{aligned}$$

with equality only when $a = b = 1/2$. So we

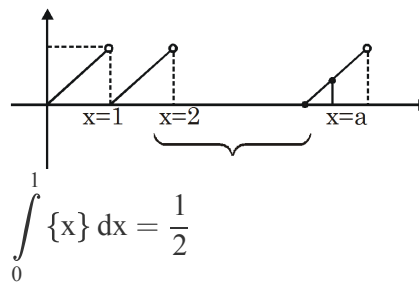
have $\sin y \geq 1$; hence $\sin y = 1$; this also implies $a = b = \frac{1}{2}$;

$$\text{therefore } \sin^2 x = \frac{1}{2}; \text{ i.e., } \sin x = \pm \frac{1}{\sqrt{2}}$$

Thus the solutions are

$$x = \frac{\pi}{4} + \frac{k\pi}{2}, \quad y = \frac{\pi}{2} + 2\ell\pi, \quad k, \ell \in \mathbb{Z}.$$

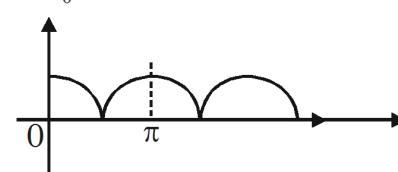
2. Ans (A)



$$\int_0^1 \{x\} dx = \frac{1}{2}$$

$$\text{so } \int_0^a \{x\} dx = 5 + \frac{1}{8} = \frac{1}{2}(10) + \frac{1}{2} \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)$$

$$\Rightarrow a = 10 + \frac{1}{2} = \frac{21}{2}$$



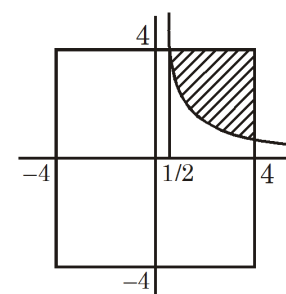
$$\text{Now } \int_0^\pi |\cos x| dx = 2 \text{ and } \int_0^{\pi/2} |\cos x| dx = 1$$

$$\begin{aligned} \Rightarrow \int_0^{b\pi} |\cos x| dx &= 11 = 10 + 1 = \int_0^{5\pi} |\cos x| dx + \int_{5\pi}^{5\pi + \pi/2} |\cos x| dx \\ &= \int_0^{11\pi/2} |\cos x| dx \Rightarrow b = \frac{11}{2} \end{aligned}$$

$$\text{so } a + b = 16$$

3. Ans (A)

$$|x| \leq 4 \text{ and } |y| \leq 4$$



Area of shaded portion

$$\Delta = \int_{1/2}^4 \frac{2}{x} dx \Rightarrow [2 \ln x]_{1/2}^4 \Rightarrow 2 [\ln 4 + \ln 2] = \ln 64$$

$$\text{Shaded region} = 14 - \ln 64$$

4. Ans (A)

$$g(x) > 0 \Rightarrow f(x) > 0$$

$$g\left(\frac{1}{2}\right) = \frac{1}{e} \Rightarrow f\left(\frac{1}{2}\right) = 1$$

$$\frac{dg(x)}{dx} < 0 \Rightarrow f'(x) < 2f(x)$$

$$\Rightarrow \frac{f'(x)}{f(x)} < 2$$

$$\Rightarrow \int_{1/2}^x \frac{f'(x)}{f(x)} dx < \int_{1/2}^x 2 dx$$

$$\therefore 0 < f(x) < e^{2x-1}$$

$$\int_{1/2}^1 0 dx < \int_{1/2}^1 f(x) dx < \int_{1/2}^1 e^{2x-1} dx$$

$$0 < \int_{1/2}^1 f(x) dx < \frac{e-1}{2}$$

5. Ans (C)

$$\cos^{-1}(x^2 - 5x + 6) = \cos^{-1}x + \sin^{-1}x$$

$$\cos^{-1}(x^2 - 5x + 6) = \frac{\pi}{2} \quad x \in [-1, 1]$$

$$x^2 - 5x + 6 = 0 \Rightarrow x = 2, 3 \text{ (reject)}$$

6. Ans (B)

$$x^5 - 1 = (x - 1)(x - \alpha_1)(x - \alpha_2)(x - \alpha_3)(x - \alpha_4)$$

$$\frac{x^5 - 1}{x - 1} = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3)(x - \alpha_4) \dots (i)$$

$$\text{at } x = \omega : \frac{\omega^5 - 1}{\omega - 1}$$

$$= (\omega - \alpha_1)(\omega - \alpha_2)(\omega - \alpha_3)(\omega - \alpha_4) \dots (ii)$$

similarly at $x = \omega^2$ in (1), we get

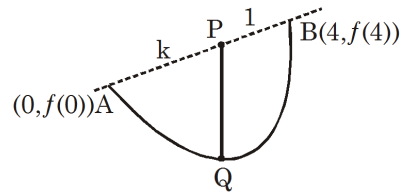
$$\frac{\omega^2 - 1}{\omega^2 - 1} = (\omega^2 - \alpha_1)(\omega^2 - \alpha_2)(\omega^2 - \alpha_3)(\omega^2 - \alpha_4) \dots (iii)$$

$$\frac{ii}{iii} \Rightarrow \prod_{i=1}^4 \frac{\omega - \alpha_i}{\omega^2 - \alpha_i} = \frac{\omega^4 + 1 - 2\omega^2}{\omega^2 + 1 - 2\omega} = \omega$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (C,D)



$f(x)$ is concave up & increasing function.

$$\frac{kf(4) + f(0)}{k+1} > f\left(\frac{4k}{k+1}\right)$$

8. Ans (C,D)

$$|A^5 - 2A^4| = 0 \text{ and } A^{-1} = \frac{1}{2}(A^2 - 5A + 7I)$$

9. Ans (A,D)

Considering n white and n black balls,

$$n(S) = {}^{2n}C_2 \cdot {}^{2n-2}C_2 \cdot {}^{2n-4}C_2 \dots {}^4C_2 \cdot {}^2C_2$$

$$n(S) = \frac{(2n)!}{2^n}$$

$$n(E) = ({}^nC_1 \cdot {}^nC_1)({}^{n-1}C_1 \cdot {}^{n-1}C_1) \dots ({}^1C_1 \cdot {}^1C_1) = (n!)^2$$

$$\therefore P = \frac{n(E)}{n(S)} = \frac{(n!)^2}{(2n)!} \cdot 2^n = \frac{2^n}{{}^{2n}C_n}$$

$$\therefore P(2) = \frac{2^2}{{}^4C_2} = \frac{2}{3}; P(6) = \frac{16}{231}$$

10. Ans (B,C,D)

$$f(x) = 13|\sin(x + \theta)|(x^2 + bx + c)$$

For $f(x)$ to be derivable in $(0, 2\pi)$;

$$x^2 + bx + c = 0$$

must have roots $\pi - \theta$ and $2\pi - \theta$.

$$(x + \theta)^2 - 3\pi(x + \theta) + 2\pi^2 = 0$$

Now compare to get

$$m = \theta = \tan^{-1}\left(\frac{12}{5}\right); n = 3 \text{ and } p = 2$$

11. Ans (B,C)

$$|\hat{a} \times (\hat{b} \times \hat{c})|^2 = |(\hat{a} \cdot \hat{c})\hat{b} - (\hat{a} \cdot \hat{b})\hat{c}|^2$$

$$\Rightarrow \sin^2(90 - \theta) \cdot \sin^2 \frac{\pi}{4}$$

$$= \left| \left(\cos \frac{\pi}{3} \right) \hat{b} - \left(\cos \frac{\pi}{6} \right) \hat{c} \right|^2$$

$$\Rightarrow \cos^2 \theta = 2 - \sqrt{\frac{3}{2}}$$

$$\text{Also, } \sin^2 \theta = 1 - \cos^2 \theta = \sqrt{\frac{3}{2}} - 1$$

12. Ans (A,B,C)

If L_1 & L_2 are coplanar then

$$\begin{vmatrix} x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = 0$$

$$\begin{vmatrix} 3 & -2 & 9 \\ 2 & -3 & 8 \\ 1 & -4 & 7 \end{vmatrix}$$

$$3(11) + 2(6) + 9(-5)$$

$$33 + 12 - 45 = 0$$

$$\text{Angle } \cos \theta = \frac{2 + 12 + 56}{\sqrt{77}\sqrt{66}} = \frac{70}{11\sqrt{42}}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (2.00)

Equation of required circle

$$x^2 + y^2 - 2x + 4y - 1 + \lambda(x^2 + y^2 + 4x - 2y - 5) = 0$$

Apply condition of orthogonality with

$$x^2 + y^2 = 4$$

$$\frac{1 + 5\lambda}{1 + \lambda} = -4$$

$$\Rightarrow 1 + 5\lambda = -4 - 4\lambda$$

$$\Rightarrow 9\lambda = -5 \Rightarrow \lambda = -\frac{5}{9}$$

$$\text{length of tangent from origin} = \sqrt{S_1}$$

$$= \sqrt{4} = 2$$

2. Ans (5.00)

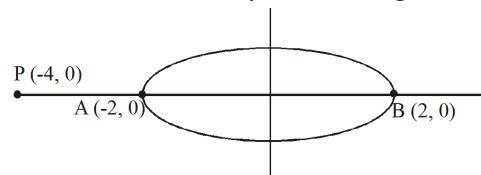
$$A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow A^n = \begin{cases} I & n \text{ is even} \\ A & n \text{ is odd} \end{cases}$$

$$\sum_{r=1}^{100} A^r = (A + A^3 + A^5 + \dots + A^{99}) + (A^2 + A^4 + \dots + A^{100})$$

$$= \sqrt{25} = 5$$

3. Ans (9.00)

$$\text{Given curve is } \frac{(x-2)^2}{4} + \frac{(y+1)^2}{1} = 1$$



$$\Rightarrow \frac{X^2}{4} + \frac{Y^2}{1} = 1$$

$$\text{Now } \frac{\max (X+4)^2 + Y^2}{\min (X+4)^2 + Y^2}$$

$$= \frac{\text{square of max distance from P to any point on the ellipse}}{\text{square of min distance from P to any point on the ellipse}}$$

$$= \frac{(PB)^2}{(PA)^2} = 9$$

4. Ans (7.00)

Let directrix touches of $(2\cos\alpha, 2\sin\alpha)$ equation of directrix is $x\cos\alpha + y\sin\alpha = 2$

Let focus (x_1, y_1) and $A(1,0)$, $B(-1,0)$ lies on parabola so

$$(x_1 - 1)^2 + y_1^2 = (\cos\alpha - 2)^2 \quad \dots(1)$$

$$(x_1 + 1)^2 + y_1^2 = (\cos\alpha + 2)^2 \quad \dots(2)$$

$$x_1 = 2\cos\alpha \quad y_1 = \sqrt{3}\sin\alpha$$

$$\text{Locus of focus } \Rightarrow \frac{x^2}{4} + \frac{y^2}{3} = 1$$

5. Ans (9.00)

$$\text{L.H.S.} = (1 + 2x^2 + x^4)(1 + C_1x + C_2x^2 + C_3x^3 + \dots)$$

$$\text{R.H.S.} = a_0 + a_1x + a_2x^2 + \dots$$

Comparing coefficient of variable we get $n = 2, 3, 4$

6. Ans (6.00)

$$f(x) = f(6-x)$$

$$f'(0) = f'(2) = f'(5) = 0$$

$$f'(x) = -f'(6-x)$$

$$f'(6) = f'(4) = f'(1) = 0 = f'(3)$$

$$h(x) = f''(x)^2 + f'(x)f'''(x)$$

$$= \frac{d}{dx}(f'(x)f''(x))$$

$$f'(x) = 0 \text{ has 7 roots}$$

$$f''(x) = 0 \text{ has min of 6 roots}$$

$$h(x) = 0 \text{ has min of 12 roots}$$

$$\boxed{n = 12}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

PRE - SCORE

17-11-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : I(A), TTAS & TEAS)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	A,B,C,D	A,C	B,D	B,D	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.33	0.50	2.43 to 2.44	1.62	50.00	75.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	4	4	4	1	2

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C,D	A,C	A,D	A,B,C,D	A,B,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	4.00	1.25	6.00	10.00	5.00	6.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	2	8	5	7	1	4

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	C,D	C,D	A,C	A,B,C,D	B,C,D	B
SECTION-II	Q.	1	2	3	4	5	6
	A.	600.00	9.00	29.00	14.00	1.00	5.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	5	6	9	5	5	2

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

- Ans (B,C)**
for $\cos \phi \Rightarrow \frac{1}{\sqrt{2}} \Rightarrow x_L = R$
for $\cos \phi = 1 \Rightarrow x_L = x_C$
 $\therefore z \Rightarrow \sqrt{R^2 + x_L^2} \Rightarrow \sqrt{2}R = \sqrt{2}x_C$
- Ans (A,B,C,D)**
 $M = \int_0^L \lambda_0 x dx = \frac{\lambda_0 L^2}{2}$
 $I_A = \int_0^L (\lambda_0 x dx) x^2 = \frac{ML^2}{2}$
 $x_{cm} \Rightarrow \frac{2L}{3}$
 $\therefore I_{COM} = \frac{ML^2}{2} - M \left(\frac{2L}{3} \right)^2$

3. Ans (A,C)

From Mosley law,

$$\sqrt{\frac{I}{\lambda}} \propto (z-1) \text{ for } K_{\alpha} \text{ line}$$

$$\therefore \frac{\lambda_z}{\lambda_1} = \left(\frac{z_{imp1} - 1}{z_{target} - 1} \right)^2 \dots\dots (i)$$

for 2nd impurity

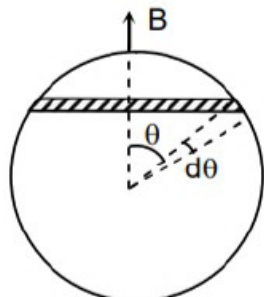
$$\frac{\lambda_z}{\lambda_2} = \left(\frac{z_{imp2} - 1}{z_{target} - 1} \right)^2 \dots\dots (ii)$$

4. Ans (B,D)

Electric potential energy 'U' = self energy + interaction energy.

$$= \frac{kQ^2}{6R} + \frac{kQ^2}{2R} - \frac{kQ^2}{3R}$$

5. Ans (B,D)

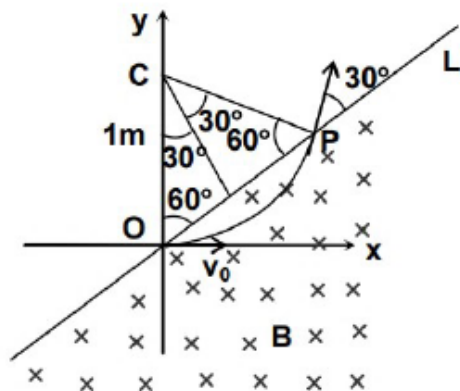


$$\varepsilon = \pi R^2 \sin^2 \theta \alpha$$

$$\text{So current } di = \frac{\varepsilon}{dr} = \frac{1}{2\rho} dR^2 \alpha \sin \theta d\theta$$

$$\text{So } \int d\mu = \int_0^\pi \frac{\pi dR^4 \alpha \sin^3 \theta d\theta}{2\rho} = \frac{2\pi}{3\rho} dR^4 \alpha$$

6. Ans (A,C,D)



$$OP = 1\text{m}$$

and time period

$$T = \frac{2\pi m}{qB} = 2\pi \text{ sec}$$

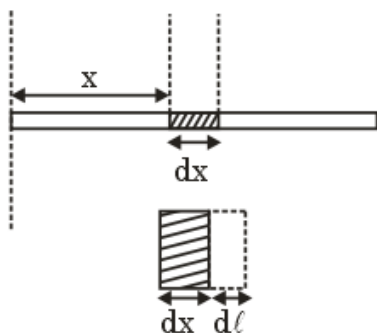
Time spent by the particle in the magnetic field is

$$t = \frac{\pi}{3\omega} = \frac{\pi m}{3qB} = \frac{\pi}{3} \text{ sec}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (0.33)



$$T(x) = \left(\frac{M\omega^2}{2L} \right) (L^2 - x^2)$$

$$Y \frac{d\ell}{dx} = \frac{T}{A}$$

$$d\ell = \frac{T}{AY} dx$$

$$\int_0^L d\ell = \left(\frac{M\omega^2}{2LAY} \right) \int_0^L (L^2 - x^2) dx$$

$$\Delta\ell = \left(\frac{M\omega^2 L^2}{3AY} \right)$$

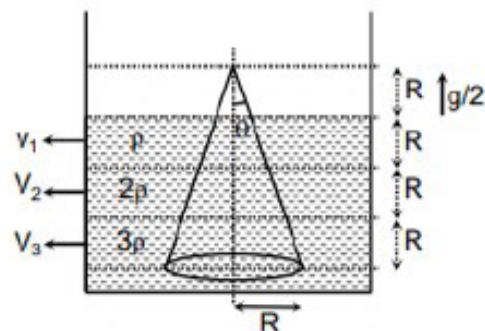
2. Ans (0.5)

$$\frac{\mu_0 i}{4\pi R \sin 30^\circ} [\sin 60^\circ] - \frac{\mu_0 i}{4\pi R \sin 60^\circ} \sin 30^\circ$$

$$\frac{\mu_0 i}{4\pi R} [\tan 60^\circ - \tan 30^\circ]$$

$$\frac{\mu_0 i}{4\pi R} \left[\sqrt{3} - \frac{1}{\sqrt{3}} \right] \Rightarrow \frac{\mu_0 i}{2\pi R \sqrt{3}}$$

3. Ans (2.43 to 2.44)



$$\tan \theta = \frac{1}{4}$$

$$v_1 = \frac{\pi}{3} [x_2^2 \cdot 2R - x_1^2 \cdot R] = \frac{7\pi}{48} R^3$$

$$v_2 = \frac{\pi}{3} [x_3^2 \cdot 3R - x_2^2 \cdot 2R] = \frac{19\pi}{48} R^3$$

$$v_3 = \frac{\pi}{3} [R^2 \cdot 4R - x_3^2 \cdot 3R] = \frac{37\pi}{48} R^3$$

$$\text{So, } m = v_1 \rho + v_2 (2\rho) + v_3 (3\rho)$$

$$\rho_0 \frac{\pi R^2}{3} \times 4R \left(\frac{3g}{2} \right) = \rho \frac{\pi R^3}{48} (156) \left(\frac{3g}{2} \right)$$

$$\text{Density of cone, } \rho_0 = \frac{39}{16} \rho$$

4. Ans (1.62)

$$\text{Path difference at 'O' } = (\mu_1 - \mu_2)t = 4.5 \times 10^{-6} \text{ m}$$

$$\text{Hence phase difference } \phi = \frac{2\pi}{\lambda} \cdot \Delta r = 20\pi$$

$$\text{Now, } I_1 = \frac{I_0}{16} \text{ and } I_2 = \frac{I_0}{25}$$

$$\text{So, } I = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$= 1.62 \text{ W/m}^2$$

5. **Ans (50.00)**

$$f_{\text{lens}} = 40 \text{ cm}$$

$$\frac{1}{f_{\text{eq}}} = \frac{1}{f_m} - \frac{2}{f_l} = \frac{1}{-20} - \frac{2}{40} = -\frac{4}{40}$$

$$\text{Distance} = 50$$

6. **Ans (75.00)**

$$3 = \frac{4}{\mu} \Rightarrow \mu = \frac{4}{3}$$

$$R = 25 \text{ cm}$$

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$f = 75 \text{ cm}$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (3)**

Displacement of A along incline will be half of displacement of B is upward direction

$$S_A = \frac{1}{2} S_B$$

$$y_A = S_A \sin 37^\circ$$

$$y_B = S_B$$

$$\frac{y_A}{y_B} \Rightarrow \frac{1}{2} \times \frac{3}{5} \Rightarrow \frac{3}{10}$$

2. **Ans (4)**

$$\psi = a \sin \left[\omega t - \frac{2\pi}{\lambda} (x \cos \alpha + y \cos \beta) \right]$$

represents a wave travelling along a line in x-y plane through origin making an angle α with x-axis and β with the y-axis.

$$\Delta \phi = \frac{2\pi}{\lambda} [(x_2 - x_1) \cos \alpha + (y_2 - y_1) \cos \beta]$$

comparing with the given equation, we get

$$\alpha = 30^\circ, \beta = 60^\circ, \lambda = 1 \text{ m}, \omega = 30/\text{s}$$

$$\text{Also, } \Delta \phi = 4\pi.$$

3. **Ans (4)**

$$X = A \cos \omega t$$

$$V = \frac{dx}{dt} = -A\omega \sin \omega t$$

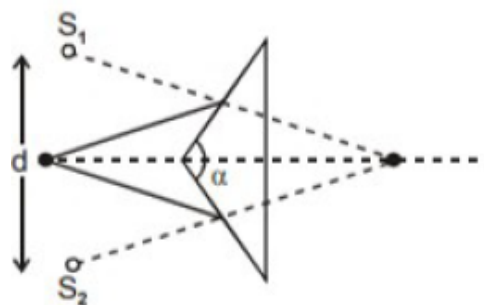
$$a = \frac{d^2x}{dt^2} = -A\omega^2 \cos \omega t$$

$$a_{\text{max}} = A\omega^2$$

$$\tau_{\text{max}} = I\alpha_{\text{max}} = I \frac{a_{\text{max}}}{R} = \frac{1}{2} MR^2 \frac{A\omega^2}{R} = \frac{1}{2} MRA\omega^2$$

$$\tau_{\text{max}} = 4 \text{ N.m}$$

4. **Ans (4)**



$$A = \frac{\pi - \alpha}{2} = \frac{1}{2} (\pi - \alpha)$$

$$\delta = (\mu - 1) A$$

$$= (\mu - 1) \frac{1}{2} (\pi - \alpha)$$

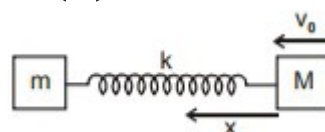
$$d = 2\ell\delta \Rightarrow d = 2\ell (\mu - 1) \cdot \frac{1}{2} (\pi - \alpha)$$

$$\Rightarrow d = (\mu - 1) \cdot (\pi - \alpha)$$

$$\Rightarrow d = (1.6 - 1) \times \frac{60}{\pi} \left(\pi - 178 \times \frac{\pi}{180} \right)$$

$$\Rightarrow 0.6 \times \frac{60}{\pi} \times \pi \times \frac{1}{90} \Rightarrow 4 \text{ mm}$$

5. **Ans (1)**



$$\mu mg = kx$$

$$x = \frac{\mu mg}{k}$$

$$w_s + w_f = k_f - k_i$$

$$-\frac{k}{2} (x^2 - 0^2) + -\mu Mgx = 0 - \frac{1}{2} mv_0^2$$

$$-\frac{k}{2} \left(\frac{\mu mg}{k} \right)^2 - \mu Mgx = -\frac{1}{2} Mv_0^2$$

$$Mv_0^2 = \frac{\mu^2 m^2 g^2}{k} + \frac{2\mu^2 Mmg^2}{k}$$

$$v_0 = \mu g \sqrt{\frac{m^2 + 2Mm}{kM}}$$

6. **Ans (2)**

Let mass of P, Q and S are m_1 , m_2 and m_3 respectively

$$\frac{Gm_1}{R^2} = \frac{G(m_1 + m_2 + m_3)}{9R^2}$$

$$8m_1 = m_2 + m_3$$

$$8\rho_1 \frac{4}{3} \pi R^3 = \frac{4}{3} \pi R^3 (7\rho_2 + 19\rho_3)$$

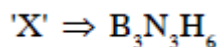
On solving

$$\frac{\rho_3}{\rho_1} = 0.2$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (B,C,D)



2. Ans (A,C)

Zr = 145 pm

Hf = 144 pm

Nb = 134 pm

Ta = 134 pm

Due to Lanthanoid contraction.

5. Ans (A,B,D)

Reaction (C) is E_{1cb}

PART-2 : CHEMISTRY

SECTION-II

3. Ans (6.00)



$$t = 0 \quad 0.8$$

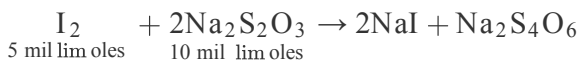
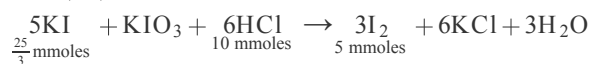
$$t = t \quad 0.8 - x$$

$$x = \frac{15.68}{22.4} = 0.7 \Rightarrow t = 3t_{1/2} = 6\text{hrs}$$

PART-2 : CHEMISTRY

SECTION-III

3. Ans (5)



$$x = \frac{25}{3} \text{ and } y = 10$$

$$3X - 2Y = 5$$

4. Ans (7)

(a), (c), (d), (f), (g), (h), (j)

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C,D)

$$f(x) = x^5 \text{sgn}(\text{sng}(x)) = \begin{cases} x^5 & ; x > 0 \\ 0 & ; x = 0 \\ -x^5 & ; x < 0 \end{cases}$$

$$\Rightarrow f'(x) = \begin{cases} 5x^4 & ; x > 0 \\ -5x^4 & ; x < 0 \end{cases}$$

$$\Rightarrow f'(0^+) = 0$$

2. Ans (C,D)

$$\text{Option (A)} f(x) = 0 \text{ \& } g(x) = 2\tan^{-1}x$$

$$\text{Option (B)} f(x) = \cos^{-1}x + \pi - \cos^{-1}x$$

$$\Rightarrow f(x) = \pi$$

$$g(x) = 2\cos^{-1}x$$

$$\text{Option (C)} f(x) = \cos x + \cos x = 2 \cos x = g(x)$$

$$\text{Option (D)} |\cos^{-1}x| = \cos^{-1}x$$

$$\Rightarrow f(x) = 2\cos^{-1}x = g(x)$$

3. Ans (A,C)

$$(1 - \sin^2x - 4\sin x + 6)(1 - \sin x) = 1 - \sin^2x$$

$$(1 - \sin x)[(-\sin^2x - 4\sin x + 7) - (1 + \sin x)] = 0$$

$$(1 - \sin x)[- \sin^2x - 5\sin x + 6] = 0$$

$$(\sin x - 1)(\sin^2x + 5\sin x - 6) = 0$$

$$(\sin x - 1)(\sin x - 1)(\sin x + 6) = 0$$

$$\sin x = 1$$

4. Ans (A,B,C,D)

(1) $^\infty$ form; use

$$\lim_{x \rightarrow 0} e^{\frac{4}{5x^2}(a \sin bx^2 + 4(1 - \cos^2x))} = e^2$$

$$\Rightarrow a \frac{\sin bx^2}{x^2} \times \frac{4}{5} + \frac{16}{5} \frac{\sin^2x}{x^2} = 2$$

$$\frac{4ab}{5} + \frac{16}{5} = 2 \Rightarrow 4ab = -6 \Rightarrow ab = -\frac{2}{3}$$

5. Ans (B,C,D)

$$\text{(A) Let } \sin^{-1}x = \theta \Rightarrow \sin \theta = x$$

$$f(x) = \sin 2\theta = 2\sin \theta \cos \theta = 2x\sqrt{1-x^2}$$

$$f(0) = 0 \Rightarrow \text{at } x = 0, 1, -1 \text{ (many-one)}$$

$$\text{(B) } f(x) + f(-x) = 0 \Rightarrow \text{odd function}$$

$$\text{(C) } g(x) = 2x, \text{ continuous in given interval}$$

$$\text{(D) } f(x) \text{ is continuous function}$$

6. Ans (B)

$$J - I = \int \left(\frac{x+y}{xy} dy - \frac{x+y}{x^2} dx \right)$$

$$= \int \left(\frac{x+y}{x^2y} \right) (xdy - ydx)$$

$$= \int \left(\frac{x+y}{y} \right) \left(\frac{xdy - ydx}{x^2} \right)$$

$$= \int \left(\frac{x}{y} + 1 \right) d \left(\frac{y}{x} \right)$$

$$g(x) = \frac{y}{x} + \ln \left(\frac{y}{x} \right) + c$$

$$\text{Put } x = 1 \Rightarrow 1 = 1 + c \Rightarrow c = 0$$

$$\therefore g(x) = \frac{y}{x} + \ln \left(\frac{y}{x} \right)$$

$$g(e) = e + 1$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (600.00)**

From skew symmetric matrix

$$a\alpha^2 + b\alpha + c = 0, a\beta^2 + b\beta + c = 0$$

$$a\gamma^2 + b\gamma + c = 0$$

$$\Rightarrow ax^2 + bx + c = 0 \text{ has three roots}$$

$$a = b = c = 0$$

$$\lambda = \begin{vmatrix} 1 & 1 & 2 \\ 3 & 4 & 1 \\ 9 & 0 & 3 \end{vmatrix} = 12 + 2(-36) = 60$$

2. **Ans (9.00)**

$$\int_0^x f(t)dt = k + f^2(x) \quad \dots(i)$$

Diff. w.r.t. x we get

$$f(x) = 2f(x) \cdot f'(x)$$

$$\Rightarrow f(x) = 0 \quad \text{or} \quad f'(x) = \frac{1}{2}$$

$$\because f(1) = \frac{7}{2} \Rightarrow f(x) = 0 \text{ is rejected}$$

$$\therefore f'(x) = \frac{1}{2} \Rightarrow f(x) = \frac{x}{2} + c$$

$$\text{Now } f(1) = \frac{7}{2}$$

$$\Rightarrow f(1) = C + \frac{1}{2} = \frac{7}{2} \Rightarrow C = 3$$

$$\Rightarrow f(x) = \frac{x}{2} + 3$$

Put in (i) to get

$$\int_0^x \left(\frac{t}{2} + 3 \right) dt = k + \frac{x^2}{4} + 9 + 3x$$

$$\left(\frac{t^2}{4} + 3t \right)_0^x = k + \frac{x^2}{4} + 9 + 3x$$

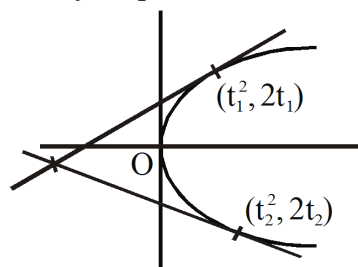
$$\frac{x^2}{4} + 3x = k + 9 + \frac{x^2}{4} + 3x$$

$$\Rightarrow k = -9$$

3. **Ans (29.00)**

$$\text{Given } \frac{2t_1}{2t_2} = \frac{4}{1}$$

$$\Rightarrow t_1 = 4t_2$$



Let point of intersection

be (h, k)

$$\therefore h = t_1 t_2 \quad k = t_1 + t_2$$

$$\Rightarrow h = 4t_2^2 \Rightarrow k = 5t_2$$

$$\therefore h = 4 \cdot \frac{k^2}{25} \Rightarrow y^2 = \frac{25}{4}x$$

$$\therefore p + q = 25 + 4 = 29$$

4. **Ans (14.00)**

$$f(1) = 0, a + b + c = 0$$

$$50 < f(7) < 60$$

$$\Rightarrow 50 < 49a + 7b + c < 60$$

$$\Rightarrow 50 < 48a + 6b < 60$$

$$\Rightarrow \frac{25}{3} < 8a + b < 10$$

$$\therefore a, b \in \mathbb{I}$$

$$\Rightarrow 8a + b = 9 \quad \dots\dots\dots(i)$$

$$\text{Similarly, } 70 < f(8) < 80$$

$$70 < 63a + 7b < 80$$

$$\Rightarrow 10 < 9a + b < \frac{80}{7}$$

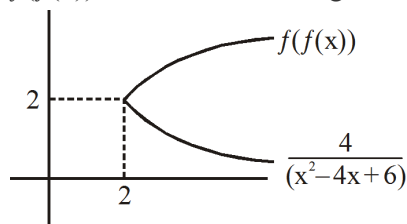
$$\Rightarrow 9a + b = 11 \quad \dots\dots\dots(ii)$$

$$\Rightarrow (a, b, c) \equiv (2, -7, 5)$$

$$\Rightarrow f(x) = 2x^2 - 7x + 5 \Rightarrow f(-1) = 14$$

5. Ans (1.00)

$f(f(x))$ is also an increasing function



$$\text{Let } y = \frac{4}{(x^2 - 4x + 6)}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{8(x-2)}{(x^2 - 4x + 6)^2}$$

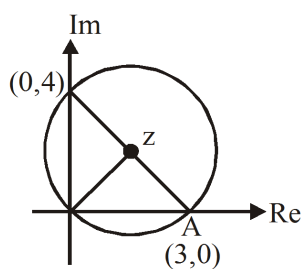
is decreasing function.

Only one solution is possible

6. Ans (5.00)

z is equidistant from $(0,0)$, $(3,0)$ & $(0,4)$

$\Rightarrow z$ is at the circumcentre of $\triangle OAB$



$$\Rightarrow |z| = \frac{5}{2}$$

$$\Rightarrow |2z| = 5$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (5)

$$\text{Let } I_1 = \int_{-5}^{-2} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 dx,$$

$$I_2 = \int_{1/6}^{1/3} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 dx$$

$$\& I_3 = \int_{6/5}^{3/2} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 dx$$

$$\text{In } I_2, \text{ put } x = \frac{1}{1-t}$$

$$\Rightarrow I_2 = \int_{-5}^{-2} \left(\frac{t^2 - t}{t^3 - 3t + 1} \right)^2 \left(\frac{1}{(1-t)^2} \right) dt$$

$$\text{In } I_3, \text{ put } x = 1 - \frac{1}{u}$$

$$\Rightarrow I_3 = \int_{-5}^{-2} \left(\frac{u^2 - u}{u^3 - 3u + 1} \right)^2 \left(\frac{1}{u^2} \right) du$$

$$\therefore I_1 + I_2 + I_3$$

$$= \int_{-5}^{-2} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 \left(1 + \frac{1}{(1-x)^2} + \frac{1}{x^2} \right) dx$$

$$\text{Let } \frac{x^3 - 3x + 1}{x^2 - x} = v$$

$$\text{or } v = x + 1 - \frac{1}{x} - \frac{1}{x-1}$$

$$dv = \left(1 + \frac{1}{x^2} + \frac{1}{(x-1)^2} \right) dx$$

$$\Rightarrow I_1 + I_2 + I_3 = \int_{-\frac{109}{30}}^{-1/6} \frac{1}{v^2} dv = -\frac{1}{v} \Bigg|_{-\frac{109}{30}}^{-1/6}$$

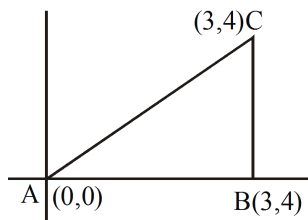
$$= - \left(-6 + \frac{30}{109} \right) = \frac{624}{109}$$

$$\Rightarrow p = 624, q = 109$$

2. **Ans (6)**

$$AC = 5 = 2a$$

$$\Rightarrow 2ae = 5 \dots (1)$$



$$\text{Also } AB + BC = 2a$$

$$\Rightarrow 2a = 7$$

$$\Rightarrow a = \frac{7}{2}$$

$$\therefore e = \frac{5}{7} \Rightarrow 1 - \frac{b^2}{a^2} = \frac{25}{49} \Rightarrow b = \sqrt{6}$$

$$\text{Area} = \pi ab = \pi \times \frac{7}{2} \times \sqrt{6}$$

$$\therefore P = 6$$

3. **Ans (9)**

$$P(R) = \frac{3}{4}, P(B) = \frac{1}{4}$$

Required probability

$$= \frac{12!}{8! 4!} \left(\frac{3}{4}\right)^4 \cdot \left(\frac{1}{4}\right)^8 \cdot \frac{3}{4}$$

4. **Ans (5)**

$$V = \begin{vmatrix} 4 & 2 \cos \theta & 2 \\ 2 & 1 & \cos \theta \\ 2 & \cos \theta & \sin \theta \end{vmatrix}$$

$$= 4(1 - \sin \theta)(1 - \cos \theta) = 2(\sqrt{2} + 1)^2$$

$$\text{when } \sin \theta = \cos \theta = -\frac{1}{\sqrt{2}}$$

$$\therefore \frac{V}{2} = (\sqrt{2} + 1)^2$$

$$\left[\frac{V}{2} \right] = 5$$

5. **Ans (5)**

$$(720 + 1)^{720}$$

$$\Rightarrow {}^{720}C_0 720^{720} + \dots + {}^{720}C_{719} 720 \cdot 1 + {}^{720}C_{720} 1^{720}$$

$$\Rightarrow \underbrace{{}^{720}C_0 720^{720} + \dots + {}^{720}C_{400} 720 \cdot 720}_{1000K} + 1$$

So we have to check last two terms ... 401

hence last 3 digits are 401 $\Rightarrow a + b + c = 5$

6. **Ans (2)**

$$(tany + y \sec^2 y) dy = e^x (\sin x + \cos x) dx$$

$$y \tan y = e^x \sin x + C$$

$$(0,0) \text{ gives } C = 0$$

$$y \tan y = e^x \sin x$$

$$\pi \tan \pi = e^{x_1} \sin x_1$$

$$\Rightarrow \sin x_1 = 0 \Rightarrow x_1 = 2\pi$$

$$\frac{x_1}{\pi} = 2$$